

DRÆM: A Discriminatively Trained Reconstruction Embedding for Surface Anomaly Detection

Scientific Report

Seminar Paper

in the context of the seminar “Advanced Machine Learning for Anomaly Detection”

at Friedrich-Alexander-Universität Erlangen-Nürnberg
at the Department Artificial Intelligence in Biomedical Engineering (AIBE)
Image Data Exploration and Analysis (IDEA) Lab

Principal Supervisor:
Associate Supervisor:

Prof. Dr. Bernhard Kainz
Johanna Müller

Author:

Christopher Roßbach
student@fau.de
123456

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My Extensions and Experiments

Adjustments to the Original Code

My work extends the code provided by the DRÆM paper (Zavrtanik et al., 2021), mainly by additional anomaly generation methods¹. To make full use of the HPC resources provided by NHR@FAU, I implemented micro-batch training and used automatic mixed precision (AMP) for the forward pass, while using full precision for loss calculation and evaluation to ensure numerical stability. In addition to reducing the memory footprint, the implementation of micro-batches fixed a bug in the original code, where the last incomplete batch of each epoch was weighted incorrectly during training (see Figure 4 in the appendix). I also switched to a cosine annealing learning rate scheduler to ensure comparability between different methods without the need for additional hyperparameter searches. These adjustments can be found in the `train_DRAEM.py` script. I performed multiple test runs after each adjustment to rule out any negative impact on the final performance. In fact, my reproduced AUROC metrics (image- and pixel-level) on the combined MVTEC dataset are within 0.1 percentage points of the values reported in the paper, while the pixel AP value improved from 68.4% to 69.7%, probably due to more stable training (average of 4 independent runs)². The scripts for running the experiments with reproducible parametrization can be found in the `slurm` directory.

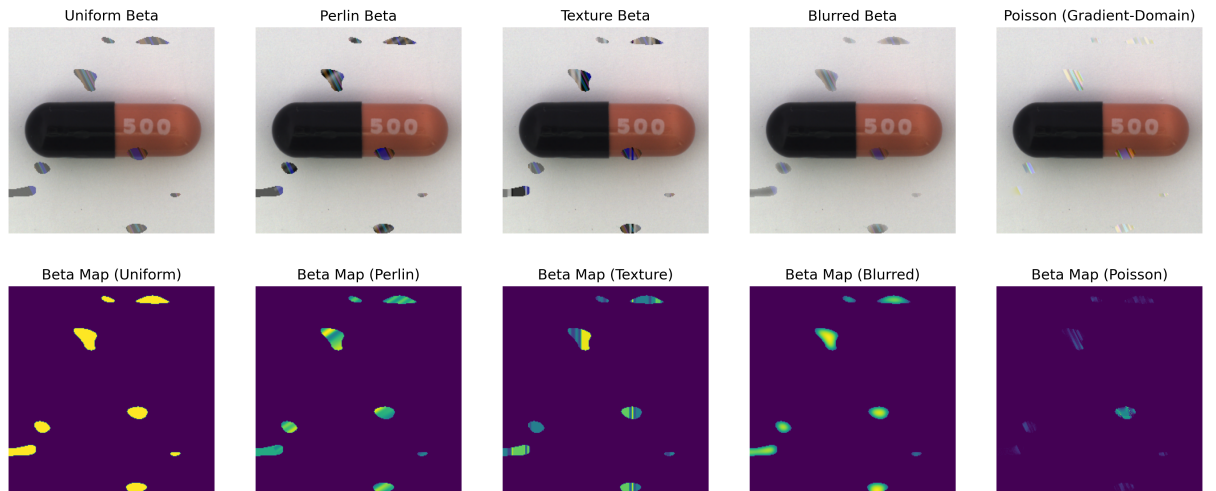


Figure 1 Different blending methods (from left to right): uniform β , Perlin- β , texture- β , blurred β , Poisson image interpolation. The top row shows an example normal image with added anomalies. The bottom row shows the strength of the contribution of the anomaly to the final image. It is clearly visible how the anomaly strength in the Poisson image interpolation is guided by the image gradients (stronger near the edge of the pill).

¹ My full code is available at https://github.com/christopher-rossbach/DRAEM_seminar_AD.

² The metrics on a per-category basis differ notably more, but this can be attributed to the natural variance of the training process. For concrete values see Figure 5.

Additional Blending Methods for Anomaly Generation

My main contribution, implementing additional blending methods for anomaly generation, is part of the data loading pipeline and thus can be found in the `data_loader.py` script. I noted that the anomaly blending method in the paper always produced anomalies with sharp borders, which are arguably rather easy to spot. By varying the anomaly strength throughout the anomaly region, more realistic anomalies can be generated, which should lead to a more robust model. For all blending methods, I kept the original anomaly generation procedure, where a random anomaly mask is created by sampling and thresholding Perlin noise patterns and then applying the anomaly in the regions defined by the mask with a strength determined by the blending method. The blending methods are: uniform β blending (as in the original paper, without spatial variation in anomaly strength), Perlin- β blending (where Perlin noise determines the anomaly opacity), texture- β blending (where the luminance of a randomly sampled texture determines the anomaly opacity), blurred β blending (where the borders of the anomaly mask are blurred), and channel-wise Poisson image interpolation, where the anomaly strength is guided by the source image gradient (see Tan et al. (2021)³). Examples of anomalies generated with these methods can be seen in Figure 1. Performance numbers for these blending methods and their combinations are reported in Table 1. As can be seen, the Poisson image interpolation, as the only content-sensitive blending method, yields on average the best results⁴. However, a closer look at the results reveals that, for some object categories, the other methods outperform the Poisson blending (see Figure 3). Combining the Poisson blending with other methods during training leverages these mutual benefits and leads to the best overall performance (last row in Table 1). The combination of methods was implemented by randomly selecting one blending method for each training image. A full excerpt of the performance metrics can be found in the appendix.

Method	Image AUROC $\uparrow$	Pixel AUROC $\uparrow$	Pixel AP $\uparrow$
DRÆM-Uniform	98.0%	97.3%	69.7%
DRÆM-Blurred	96.4%	96.0%	65.8%
DRÆM-Texture	98.0%	97.2%	69.1%
DRÆM-Perlin	98.2%	97.3%	69.5%
DRÆM-Poisson	<u>99.1%</u>	<u>97.8%</u>	<u>71.6%</u>
DRÆM-Poisson+Perlin	99.2%	98.0%	73.6%

Table 1 Key performance metrics (image-level AUROC, pixel-level AUROC, pixel-level AP) on the combined MVTec dataset for different blending methods used.

³ Tan et al. (2021) also provides the interpolation code.

⁴ This method was suggested by my supervisor, Johanna Müller.

References

- Tan, J., Hou, B., Day, T., Simpson, J., Rueckert, D., & Kainz, B. (2021). Detecting Outliers with Poisson Image Interpolation. In M. de Bruijne, P. C. Cattin, S. Cotin, N. Padoy, S. Speidel, Y. Zheng, & C. Essert (Eds.), *Medical Image Computing and Computer Assisted Intervention – MICCAI 2021* (pp. 581–591). Springer International Publishing. https://doi.org/10.1007/978-3-030-87240-3_56
- Zavrtanik, V., Kristan, M., & Skocaj, D. (2021). Draem - a discriminatively trained reconstruction embedding for surface anomaly detection. *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, 8330–8339.

Appendices

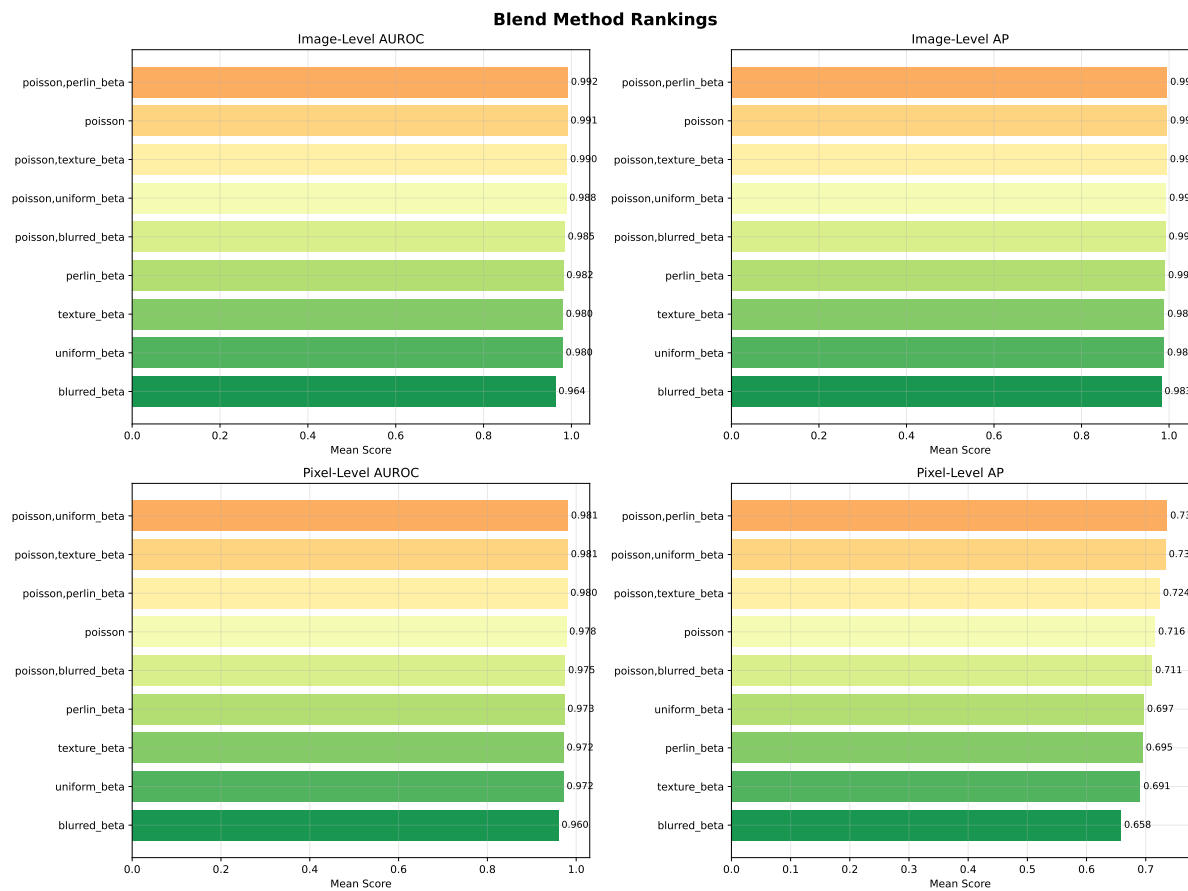


Figure 2 Combined performance comparison of blending methods and blending method combinations. The reported numbers are averages over four independent runs.

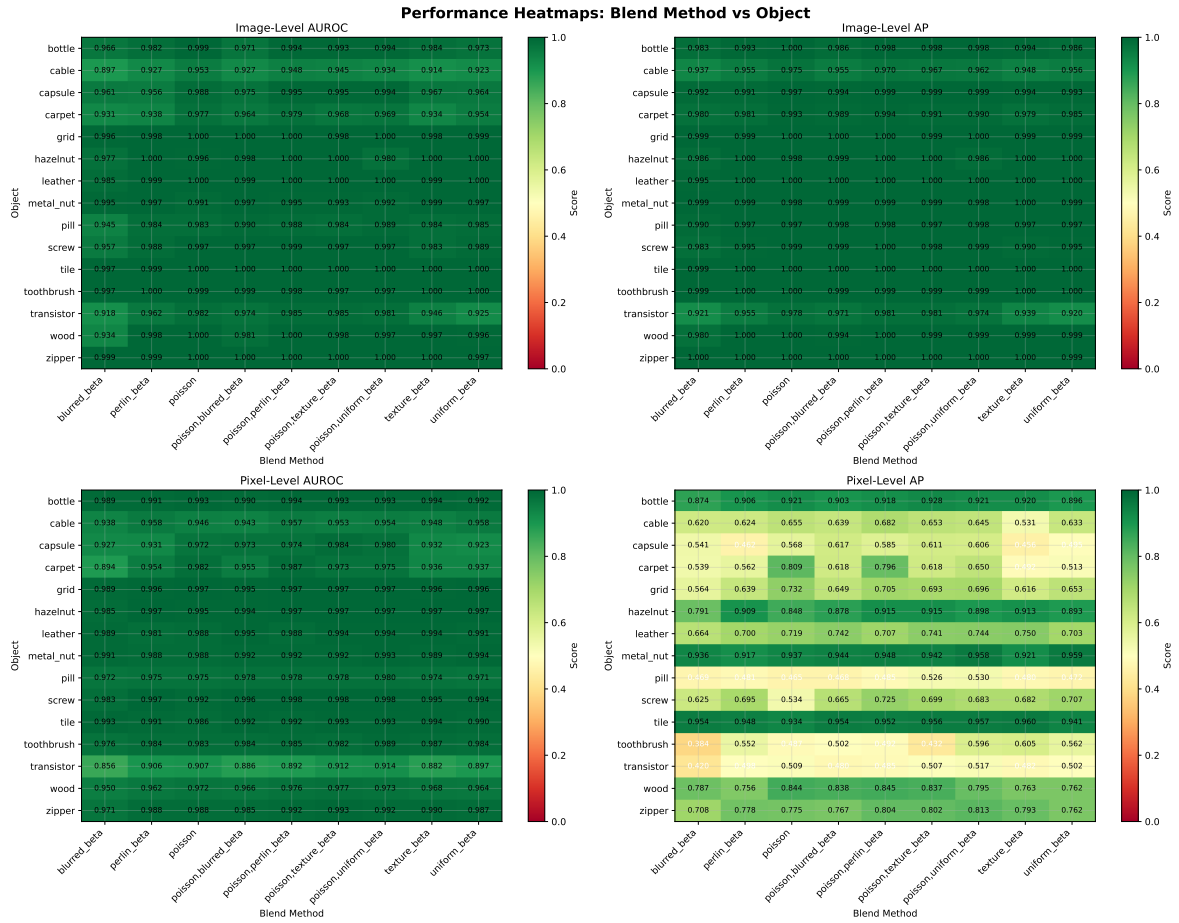


Figure 3 Detailed performance comparison of blending methods per object category. Despite performing best on average, the Poisson image interpolation (PII) blending is outperformed by other methods on some categories in terms of pixel-level AP (e.g., screw or toothbrush). The reported numbers are averages over four independent runs.



Figure 4 An example of the achieved loss smoothing by weighting the last batch of each epoch. The graph shows the SSIM loss on the training set over 700 epochs with and without correct weighting of the last incomplete batch.

Uniform vs Poisson+Perlin: Performance by Object Type

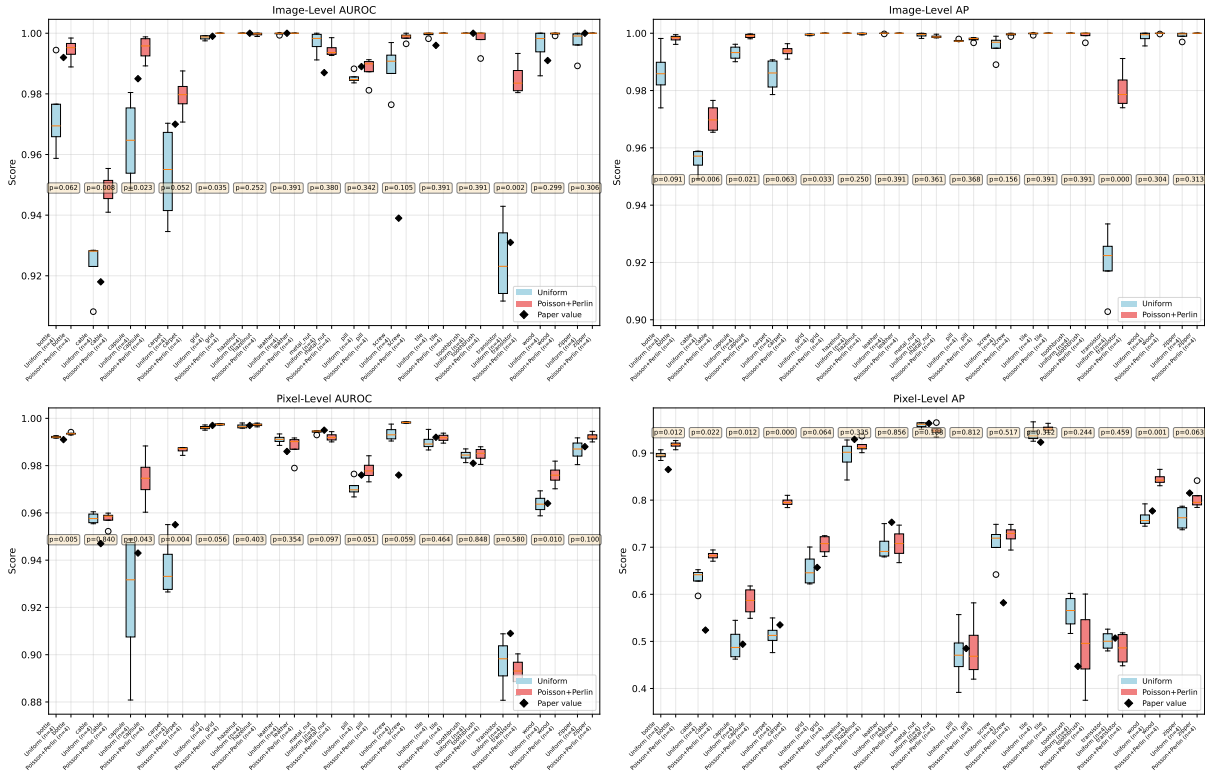


Figure 5 Detailed performance comparison of blending methods per object category. It can be seen that the combination of Poisson interpolation and Perlin- β blending outperforms the original blending method not only in general but also separately in each object category with a few exceptions. The black diamonds indicate the values that the DRÆM paper reports for the different metrics. It is notable that my experiments have large deviations within a single object category, and the paper values are often not within the spread of my results. p-values are provided for the significance of the difference between uniform- β and Poisson blending. (Note: the paper does not report image-level AP values.)

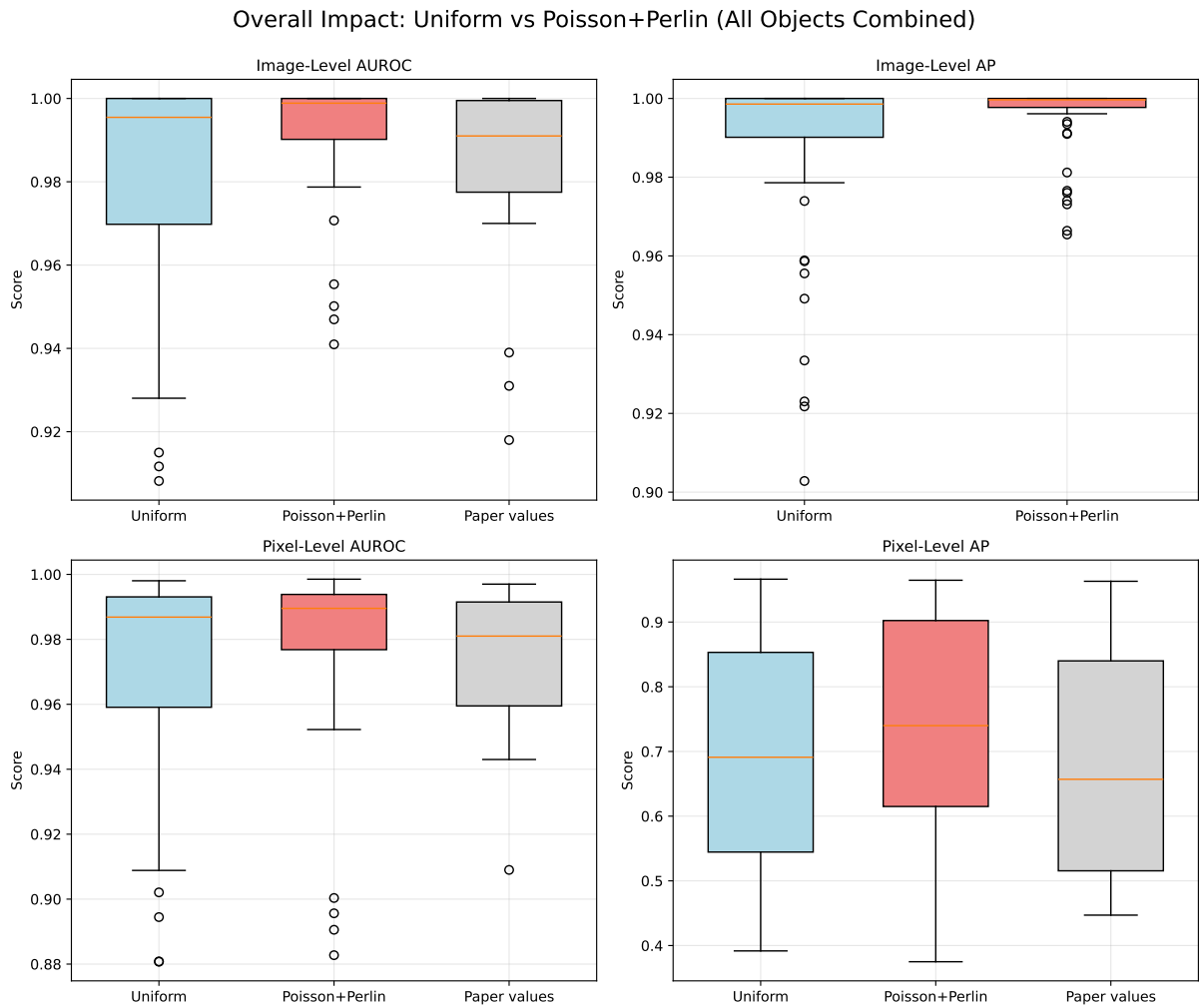


Figure 6 An aggregated version of the performance comparison in Figure 5 of blending methods over all object categories. (Note: the paper does not report image-level AP values.)

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Nürnberg, 13th January 2026

Christopher Roßbach